

CodeAlpha Biotechnology Internship

Task 2: Ginkgo Bioworks Case Study

Text-only version prepared from the original presentation PDF.

Page 1

CODEALPHABIOTECHNOLOGYINTERNSHIP

ProgrammingBiology:

HowGinkgoBioworksIs

DisruptingIndustrial

Biotechnology

CaseStudyAnalysisMeenaVigneshMInternID:CA/DF1/86669Domain:

Biotechnology

Page 2

COMPANYOVERVIEW

BiologyasSoftware

0 1

Makebiologyeasiertoengineer

-intentionally,notbyaccident.

BornatMIT,builttoindustrializesyntheticbiologyatscale.

Applyhardwareandsoftware

engineeringprinciplesto

biologicaldesign.

FoundedbyMITscientists,GinkgoBioworkstreatsDNAlikecode.If

computersrunonbinary(and),organismsrunongeneticsoftware(

A , T , C , G) .

Origin

MissionCorePhilosophy

Page 3

CORE INNOVATION

The Foundry: Automation at Biological

Scale

What It Is

Why It Matters

Ginkgo can design, build, and test

thousands of engineered organisms

simultaneously. Every experiment feeds

data back into an automated pipeline -

transforming biological trial-and-error

into a predictable engineering cycle.

Ahighly automated robotic facility that

replaces manual pipetting and traditional

bench work with custom robotics and

high-throughput liquid handlers.

Page 4

CORE TECHNOLOGY

The Codebase: Biology's GitHub

Reuse & Speed

A Proprietary Genetic Library

Continuous Learning

Engineers pull optimized sequences to assemble new organism profiles rapidly.

The Codebase is Ginkgo's massive inventory of characterized genetic parts, metabolic pathways, and engineered chassis strains - continuously growing with every project.

Each project expands the library, reducing cost and time for future cycles.

Page 5

APPLICATIONS

From Factories to Fields

Industrial Biotech

Sustainable Agriculture

Engineering specialized yeast and bacteria to produce high-value chemicals, rare fragrances, and industrial enzymes - sustainably and at scale.

Through JoynBio (Bayer partnership), engineering self-fertilizing microbes that fix atmospheric nitrogen - targeting elimination of carbon-heavy synthetic fertilizers.

Page 6

APPLICATIONS

Medicine & Biosecurity

Pharma & Vaccines

Ginkgo Concentric

Optimizing cell lines for complex biologic drugs and accelerating mRNA vaccine manufacturing efficiency at industrial scale.

A dedicated biosecurity platform operating a global pathogen monitoring network - tracking infectious diseases, biological threats, and variant mutations in real-time.

Page 7

BUSINESS MODEL

Biology-as-a-Service

Platform, Not Product

Ginkgo does not manufacture end-consumer products. It operates strictly as an R&D platform provider, lowering the barrier to entry for biotech startups that cannot afford to build their own laboratories.

Customers hire Ginkgo to design a custom microbe.

Ginkgo retains platform IP and earns through milestone payments, royalties, or equity - aligning incentives with customer success.

Page 8

CHALLENGES

Engineering Biology Is Hard

Metabolic Burden

Forcing host cells to overproduce foreign compounds causes severe metabolic stress, reducing viability and yield.

Genetic Instability

In large bioreactors, engineered organisms frequently mutate across generations, shutting down modified pathways to conserve energy.

Scale-Up Bottlenecks

A pathway that performs flawlessly in a microtiter plate often fails when scaled to a 50,000-liter industrial fermenter due to gradient differences.

Page 9

CHALLENGES

Regulatory & Market Headwinds

Public Perception

Navigating widespread skepticism and strict regional regulations surrounding GMOs and synthetic biology.

High Upfront Costs

Biosafety & Dual-Use

Risks

Ensuring high-throughput genetic design platforms cannot be misused to engineer harmful bioweapons.

Operating automated foundries demands massive capital expenditure long before generating profitable commercial royalties.

Page 10

FUTURE OUTLOOK

The AI-Augmented

Bioeconomy

The New Standard

Generative AI for

Proteins

Autonomous

Foundries

LLMstrained on biological

sequences design novel

proteins and metabolic

pathways that do not exist in

nature.

By combining The Foundry with The Codebase, Ginkgo has set the benchmark for a scalable, data-driven bioeconomy.

AI models analyze foundry data

and immediately design the

next genetic iteration - fully

closed-loop, no human

intervention.